Visualization of DE

Levels of QC

1) RNA quality (undegraded and of sufficient quantity)

2) Sequencing quality

3) Post-processing QC (today's lecture)
   - Sequence quality (mapping %, coverage)
   - PCA/MDS Plots
   - Replicate agreement

Sequence Quality

1. Fraction of mapped reads
   generally >80% for single-end and >90% for paired end

2. Number of reads
   ~20M (single-end) reads per sample for RNA-seq is ideal

3. Percentage of genes with low read counts in each sample (e.g., 1 CPM or 10 reads)

even when the mean coverage is high, many transcripts remain undetected

MDS/PCA Plots

Multi-Dimensional Scaling or Principal Component Analysis

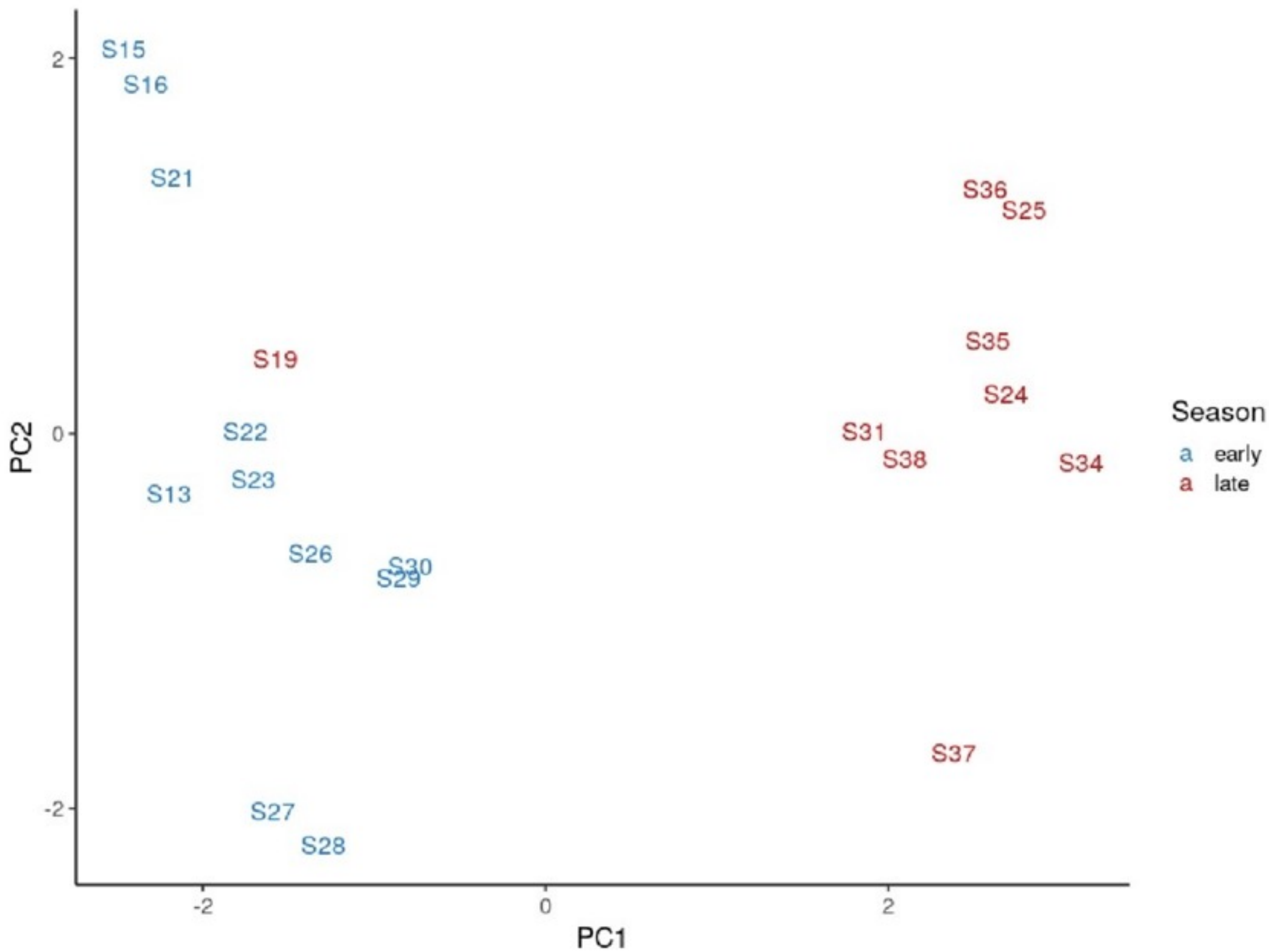
- for visualizing similarities between datasets

- translates data into pairwise distances and separates on the number of use-specified dimensions (usually two)

- the first dimension explains the largest proportion of variation across the samples, the second dimension explains the second most, and so on.

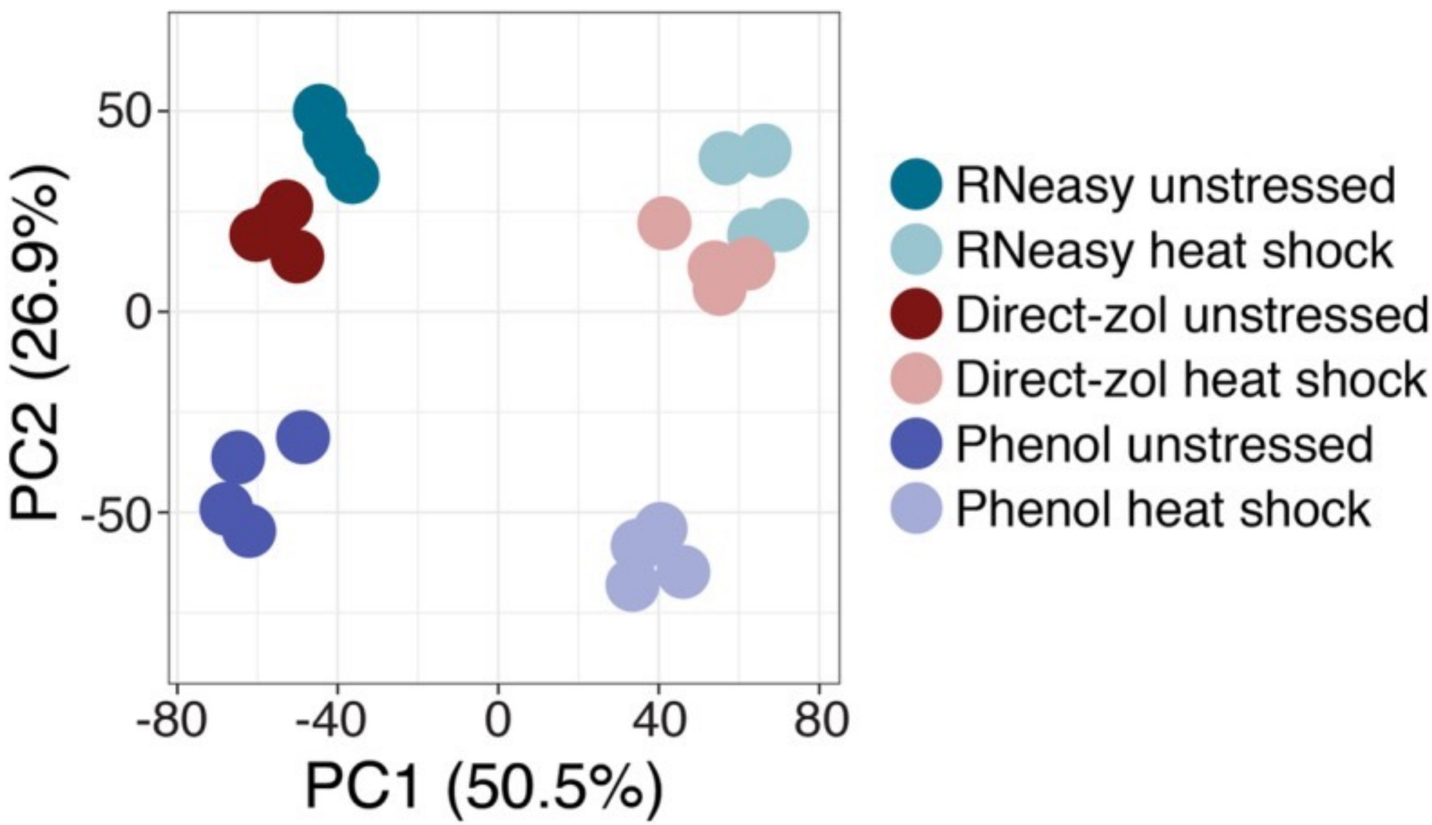
MDS/PCA Plots

Can be used to identify sample mixups



MDS/PCA Plots

Or, can tell you interesting things about your data



https://bmcgenomics.biomedcentral.com/articles/10.1186/s12864-020-6673-2

MDS/PCA Plots

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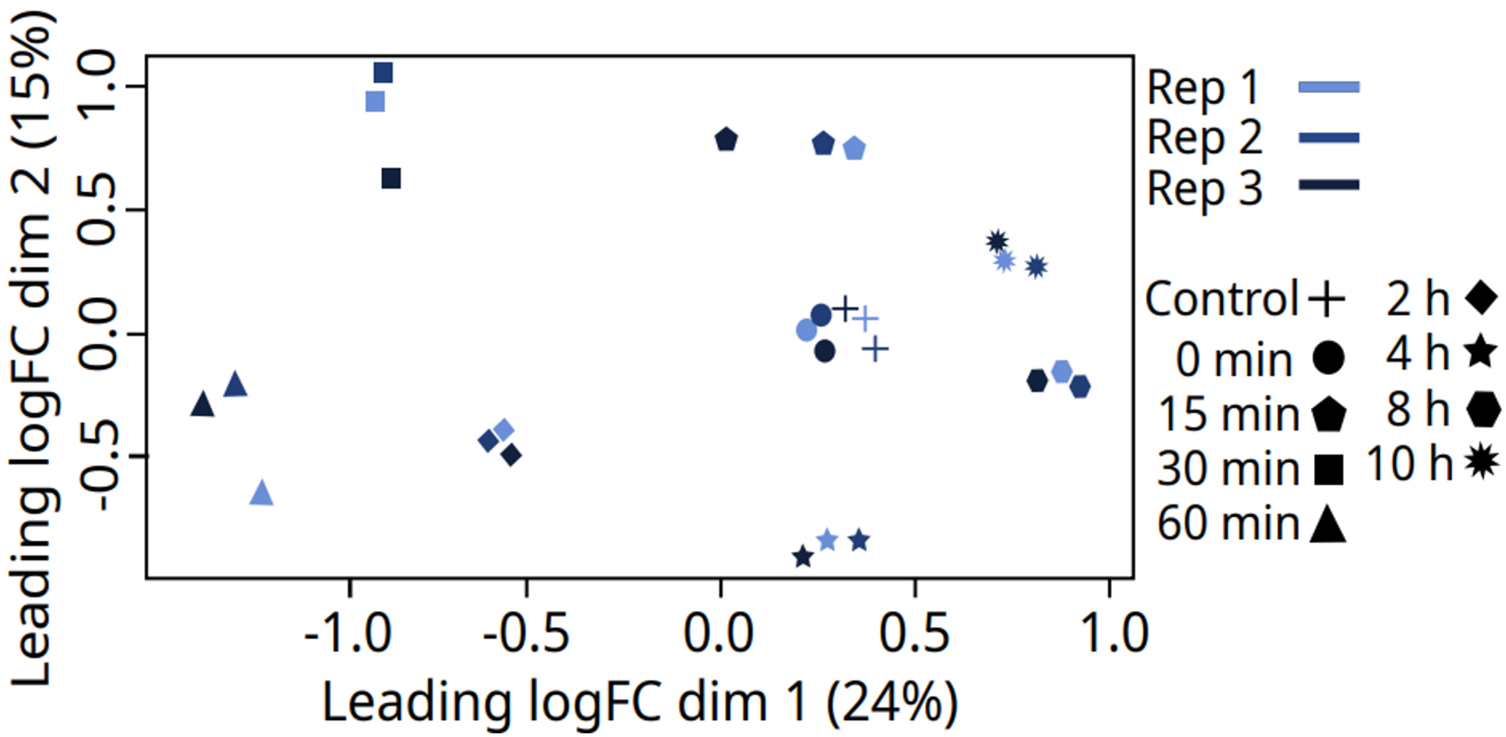
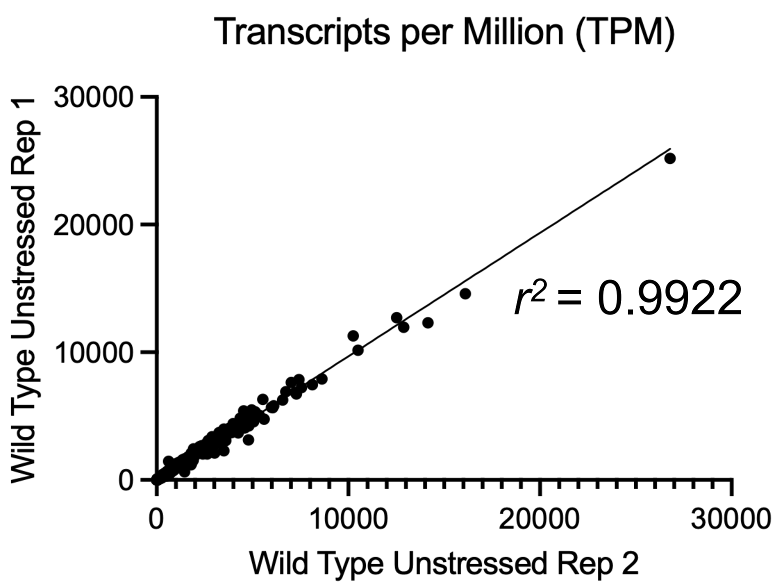
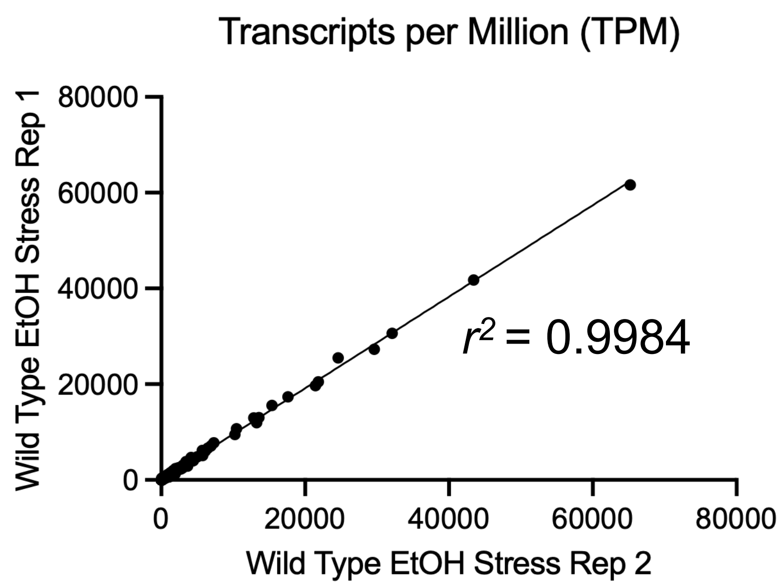


https://onlinelibrary.wiley.com/doi/10.1111/mec.16703

Replicate Agreement

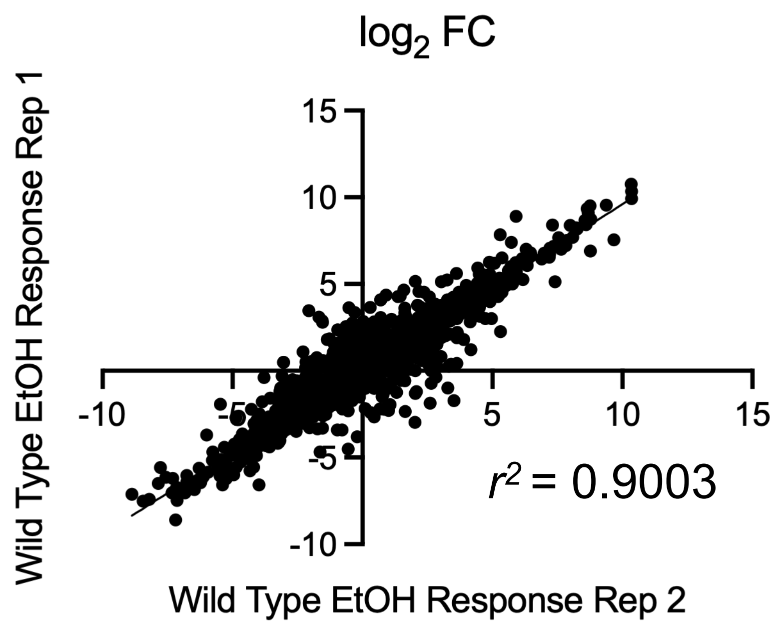
TPMs between replicates should have very good agreement





Replicate Agreement

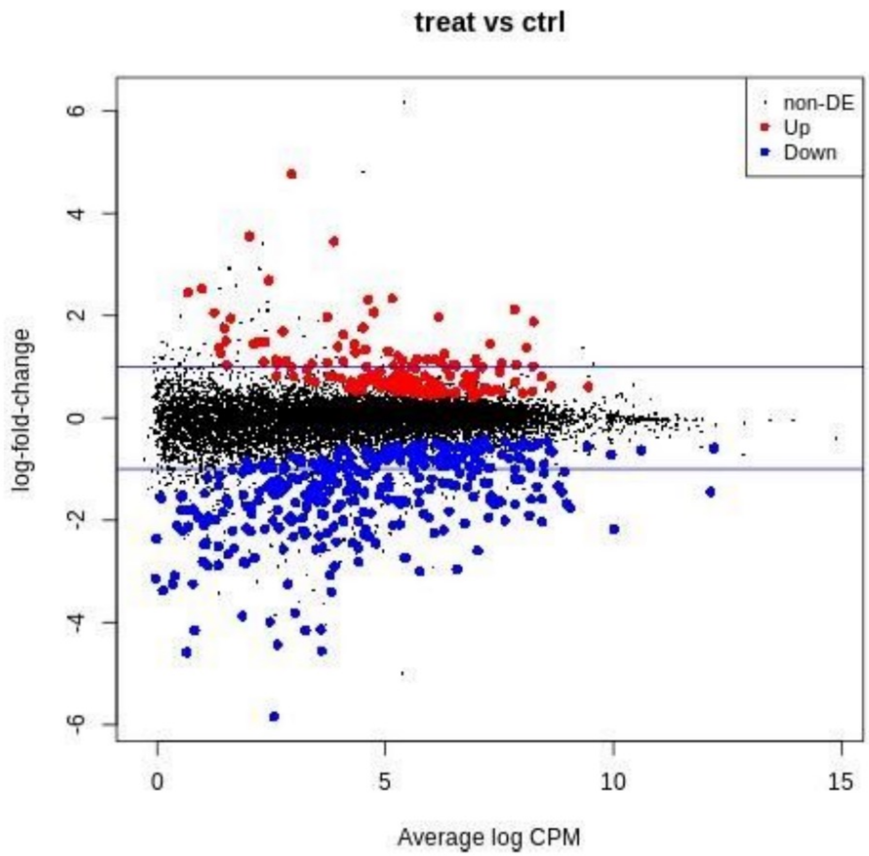
logFC between replicates tend to have less agreement, but should still be fairly high for a strong response (with lots of DE genes)



MA Plots

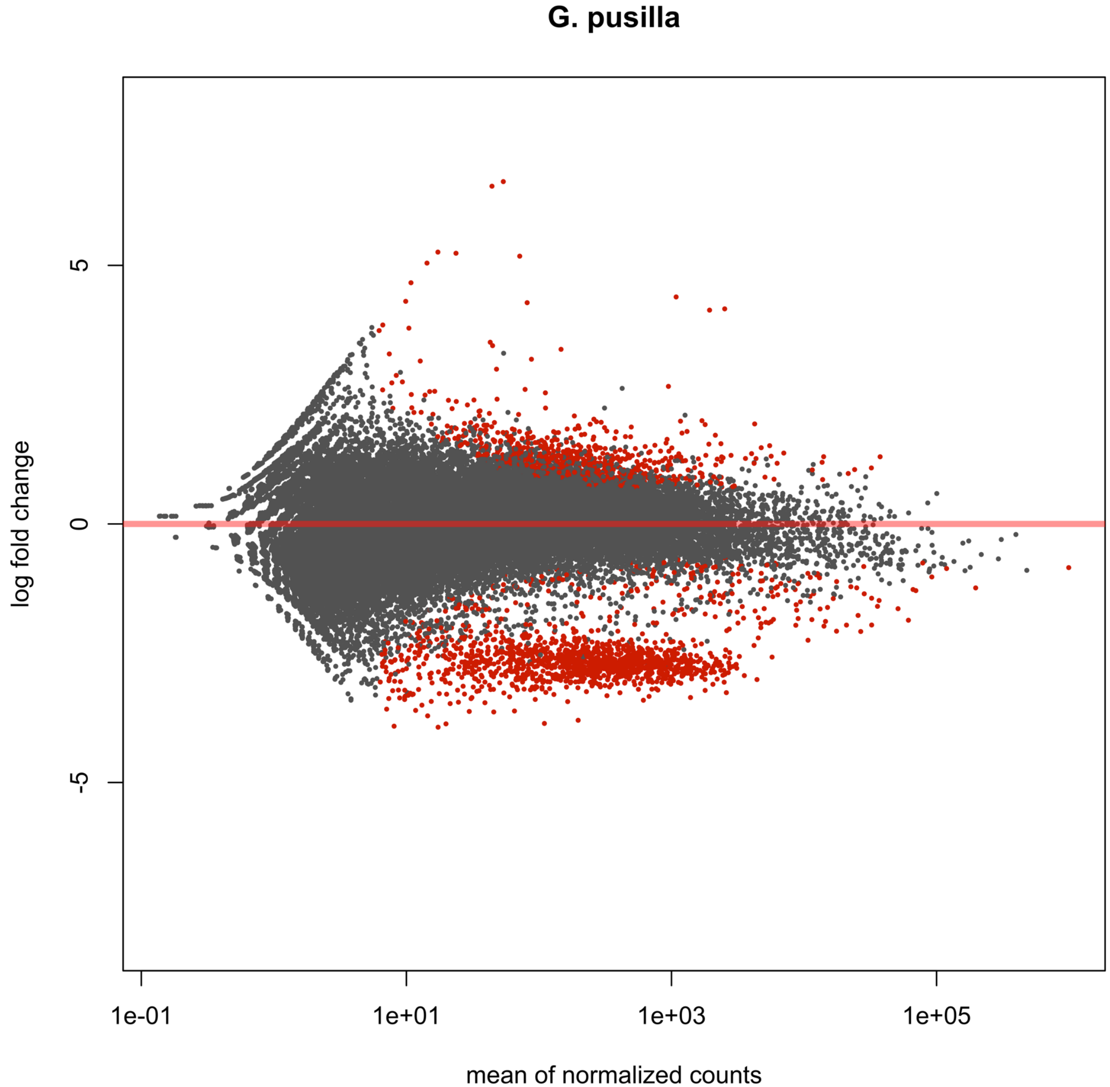
MA plots display a log ratio (M) vs an average (A)

- Plots mean normalized counts (e.g., log CPM) on the x-axis and logFC on the y-axis
- generally color coded for up/down and significance



Weird MA Plots

tissue specific expression?



Weird MA Plots

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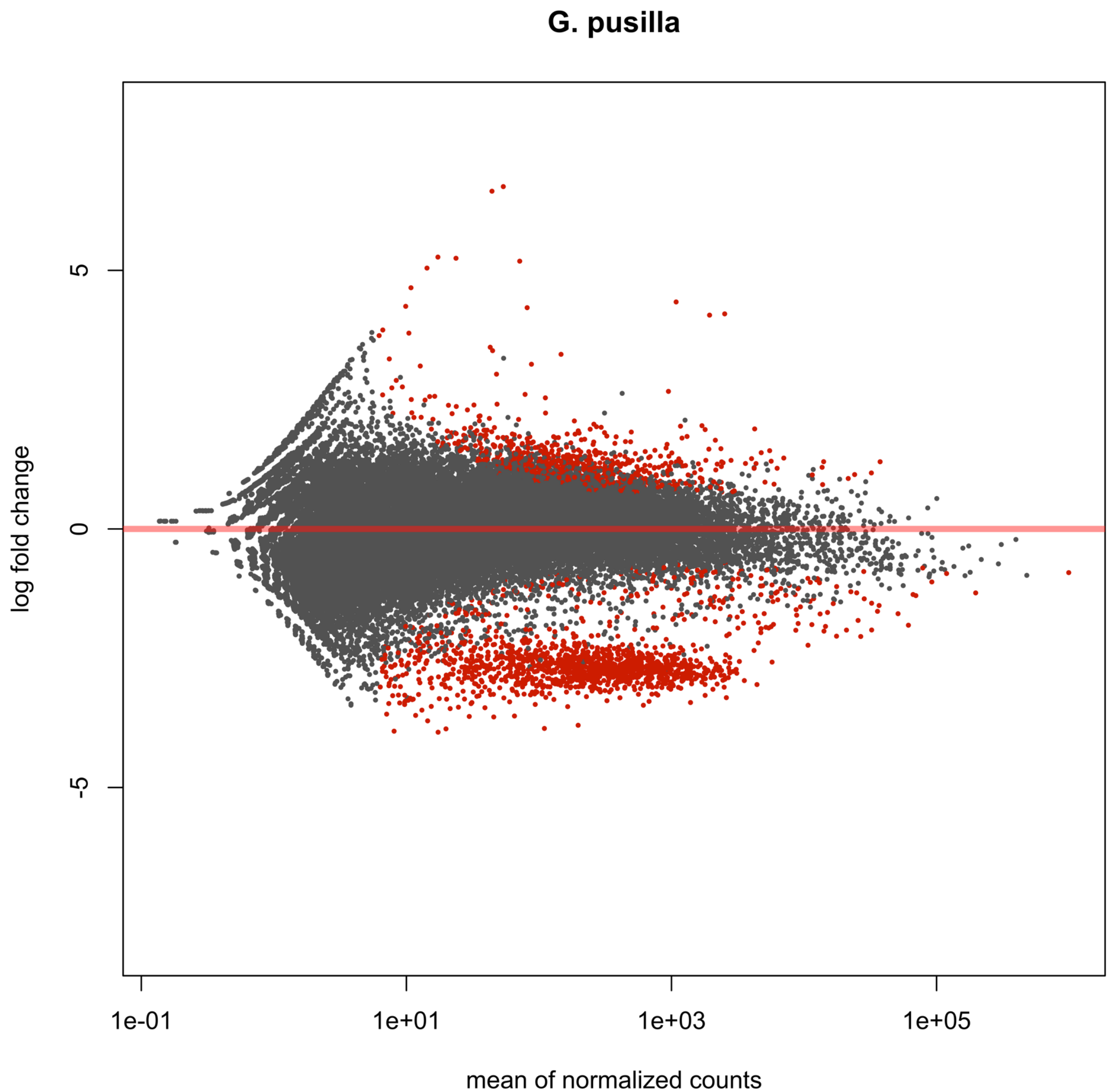
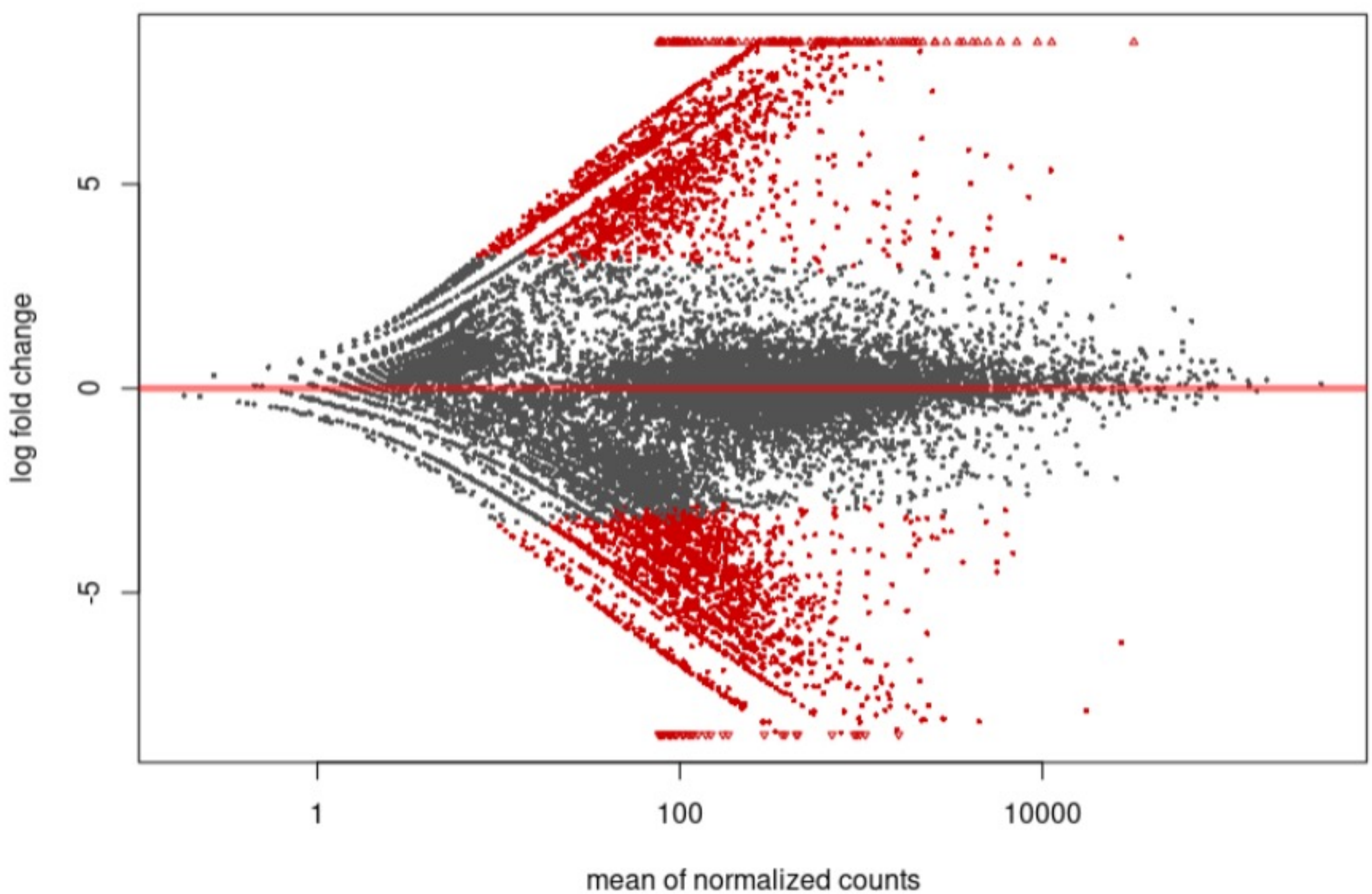


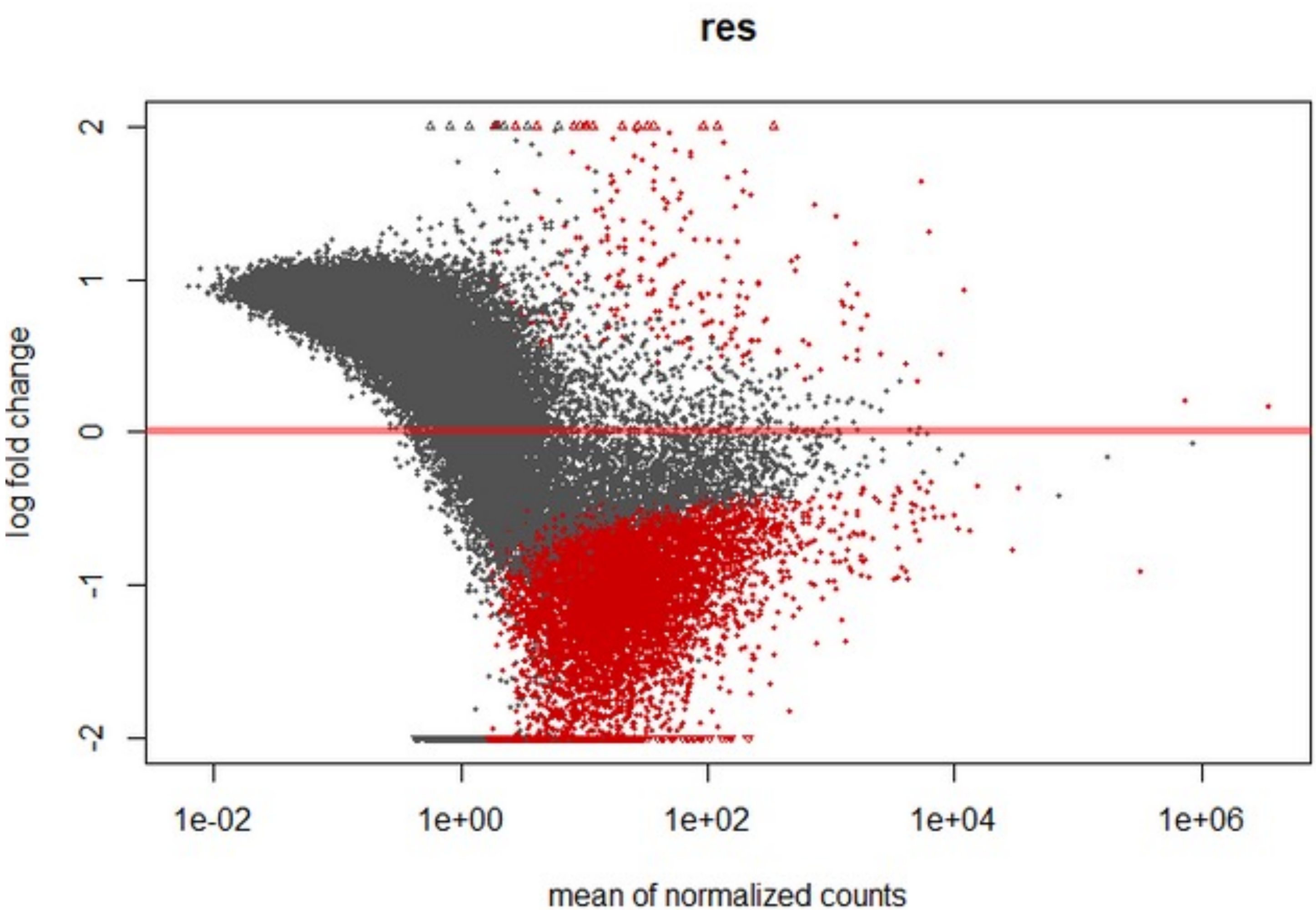
Weird MA Plots

lots of zero counts in specific samples (which affects logFC estimates)



Weird MA Plots

no %$&^%# idea, but it ain't good



Volcano Plots

Alternative to MA plots

- Plots logFC on the x-axis and –log(p-values) on the y-axis
- generally color coded for up/down and significance